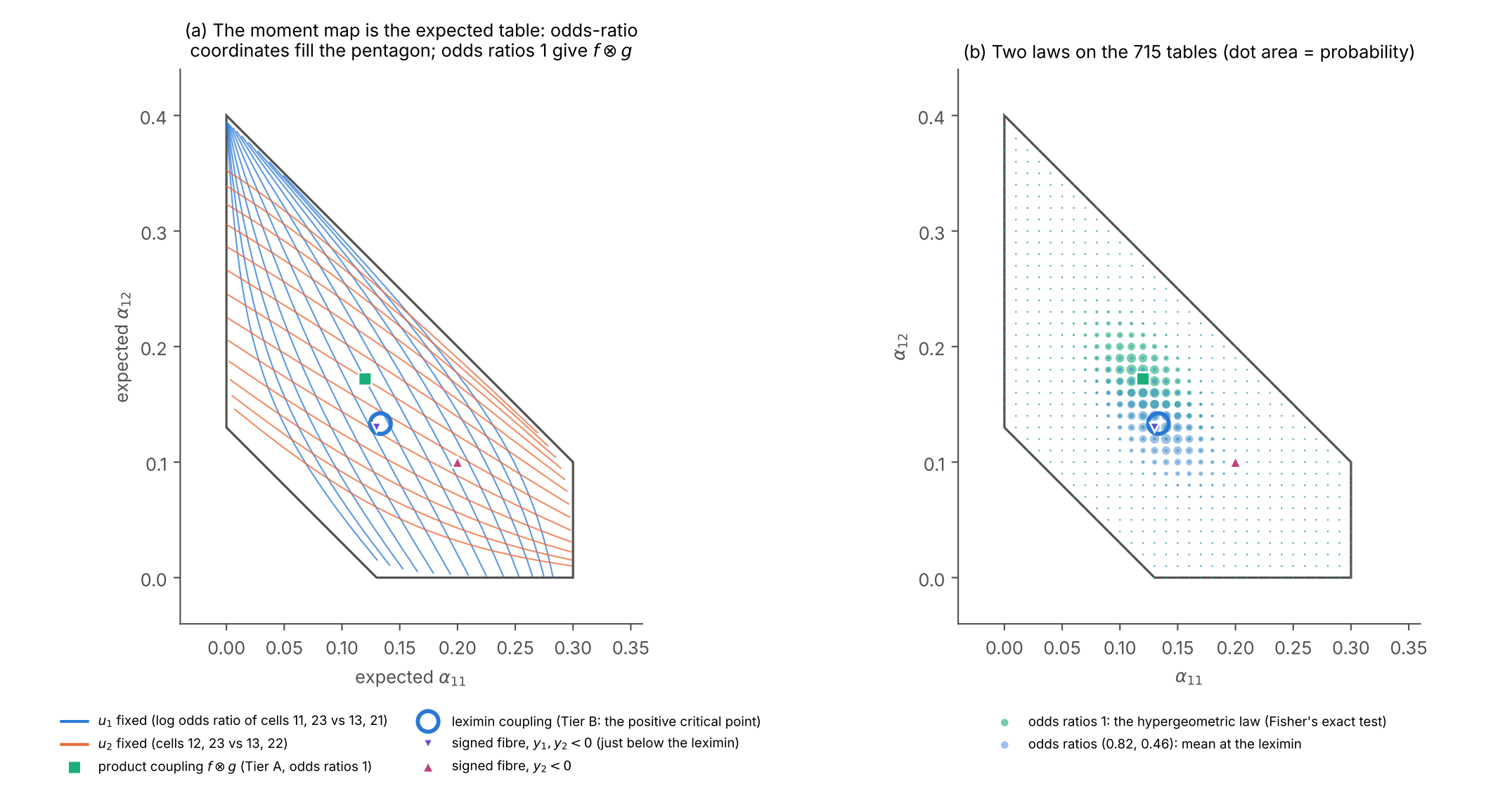

(a) The moment map is the expected table: odds-ratio coordinates fill the pentagon; odds ratios 1 give $f \otimes g$
expected α_{12}
expected α_{11}
0.0
0.1
0.2
0.3
0.4
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
u_1 fixed (log odds ratio of cells 11, 23 vs 13, 21)
u_2 fixed (cells 12, 23 vs 13, 22)
product coupling $f \otimes g$ (Tier A, odds ratios 1)
leximin coupling (Tier B: the positive critical point)
signed fibre, $y_1, y_2 < 0$ (just below the leximin)
signed fibre, $y_2 < 0$
(b) Two laws on the 715 tables (dot area = probability)
α_{12}
α_{11}
odds ratios 1: the hypergeometric law (Fisher's exact test)
odds ratios (0.82, 0.46): mean at the leximin